Note

In the present chapter, we consider bounded functions $f(x)$ that are continuous on $[a,b]$. However, the **Riemann integral** of $f(x)$ on $[a,b]$ also exists for bounded functions $f(x)$ that are piece-wise continuous on $[a,b]$. We have used the same symbol $\int$ both for definite integral and anti-derivative (indefinite integral). The reason will be clear after we state the Fundamental Theorems of Integral Calculus. The variable x is **dummy** in the sense that it is selected at our choice only. So we can write $\int_a^b f(x)dx$ as $\int_a^b f(u)du$. So, we have $\int_a^b f(x)dx = \int_a^b f(u)du$. As $\max\left(x_i - x_{i-1}\right) \to 0$, all the three points x_{i-1}, ξ_i, and x_i of each subinterval $[x_{i-1}, x_i]$ are dragged into a single point. We have already indicated that there are infinitely many ways of choosing the evaluation point ξ_i in the subinterval $[x_{i-1}, x_i]$, $i = 1, 2, \cdots, n$. By choosing $\xi_i = x_{i-1}$, $i = 1, 2, \cdots, n$, we have

$$\int_a^b f(x)dx = \lim_{n\to\infty \text{ and } \max(x_i - x_{i-1})\to 0} \sum_{i=1}^{n} f(x_{i-1})(x_i - x_{i-1}). \qquad ...(2)$$

Equation (2) is known as the **left-end rule** for evaluating the Riemann integral.

By choosing $\xi_i = x_i$, $i = 1, 2, \cdots, n$, we have

$$\int_a^b f(x)dx = \lim_{n\to\infty \text{ and } \max(x_i - x_{i-1})\to 0} \sum_{i=1}^{n} f(x_i)(x_i - x_{i-1}). \qquad ...(3)$$

Equation (3) is known as the **right-end rule** for evaluating the Riemann integral.

By choosing $\xi_i = \frac{x_{i-1} + x_i}{2}$, $i = 1, 2, \cdots, n$, we have

$$\int_a^b f(x)dx = \lim_{n\to\infty \text{ and } \max(x_i - x_{i-1})\to 0} \sum_{i=1}^{n} f\left(\frac{x_{i-1} + x_i}{2}\right)(x_i - x_{i-1}). \qquad ...(4)$$

Equation (4) is known as the **mid-point rule** for evaluating the Riemann integral.

Remarks

(1) If the Riemann integral $\int_a^b f(x)dx$ exists, then the Riemann integral $\int_a^x f(u)du$ is a well-defined real number for every $x \in [a,b]$. So, we can define a function $F(x)$ on $[a,b]$ such that $F(x) = \int_a^x f(u)du, x \in [a,b]$.

(2) If $f(x) \geq 0$ for all $x \in [a,b]$, then the Riemann integral $\int_a^b f(x)dx$ is equal to the geometric area of the region bounded by the graph of $y = f(x)$, the x-axis, the lines $x = a$ and $x = b$. See Fig. 9.3.

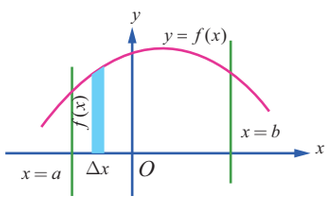


Fig. 9.3

Chapter 9 Applications of Integration.indd 93 17-02-2020 17:03:26